

Aliah University
SPRING SEMESTER EXAMINATION 2024

FOR BCA 1ST YEAR 2ND SEMESTER

Paper Name: Object Oriented Programming
Paper Code: BCAUGMIN1201

Full Marks: 80
Time: 3:00 Hrs

Section A

(Answer all questions)

5 × 2

1. (a) Object means
 - A. template of a class
 - ☒ B. instance of a class
 - C. instance variable of a class
 - D. description of a class
- (b) Size of char in Java is
 - A. 8 bits
 - ☒ B. 16 bits
 - C. 32 bits
 - D. 64 bits
- (c) The built-in class `BufferedReader` is defined in the package
 - ☒ A. `java.io`
 - B. `java.lang`
 - C. `java.awt`
 - D. None of these
- (d) Assume that the main thread runs for 10 seconds and its 2 child threads run for 5 seconds each. What is the total execution time?
 - ☒ A. 10 seconds
 - B. 15 seconds
 - C. 20 seconds
 - D. 25 seconds
- (e) Which of the following is not true about applet?
 - A. Applet runs as a part of a web document
 - B. Applet has limited access to resources
 - C. Applet runs without risk of viruses
 - D. Applet runs on a server and send the outputs

Section B

(Answer any 6 questions)

6 × 5

- ☒ 2. What is the difference between interpretation and execution? What is bytecode? Is it executed or interpreted? 2+1+2
- ☒ 3. Demonstrate the differences between constructor and method (with suitable codes). 5
4. What are the different access specifiers in Java? What happens when no access specifier is given? 3+2
5. What is the significance of static variable and static method? How can we get the number of characters of a string? 4+1
6. How to call superclass' constructor? What are the benefits of inheritance? 2+3

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| 7. Write down the differences between interface and class. What is the usefulness of interface? | 3+2 |
| 8. What are the benefits of multithreading? What is meant by context switch? | 3+2 |
| 9. Write down the advantages of applets. Mention the methods of the built-in Applet class. | 3+2 |

Section C

4 × 10

(Answer any 4 questions)

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| 10. (a) Give a comparative view of C++ and Java.
(b) Write down the principles of object-oriented programming.
(c) Why is the main method always public and static? | 3+5+2 |
| 11. (a) What is meant by call by reference?
(b) Write a Java program to calculate the factorial of a number. | 2+8 |
| 12. (a) How can one prevent inheritance?
(b) Write a Java program to demonstrate method overriding. | 2+8 |
| 13. (a) What is an exception?
(b) Write a short Java program to demonstrate Array Index Out of Bounds exception.
(c) How can a programmer throw an exception intentionally? | 2+5+3 |
| 14. (a) How are threads created?
(b) What is meant by synchronization problem? Give example.
(c) With suitable codes, show how it is solved in Java. | 2+3+5 |
| 15. Write short notes on
(a) Polymorphism
(b) Interthread communication | 5+5 |